



2025

DATA SCIENCE FOR E-COMMERCE INSIGHT

OPTIMIZING BUSINESS WITH REAL-WORLD TRANSACTIONAL DATA FROM OLIST (BRAZIL)





BUSINESS CHALLENGES IN E-COMMERCE

E-commerce platforms generate massive volumes of transactional data, yet much of it remains underutilized.

Common challenges include:

- Unidentified high-value customers
- Lack of demand forecasting
- Poor delivery analysis and review impact tracking

LETS GET STARTED





BUSINESS PROBLEMS

Existing customer and sales data have not yet been transformed into actionable business strategies.

- There is no customer segmentation in place.
- There is no product demand forecasting.
- Seller performance and delivery time cannot be effectively analyzed

Lack of Insight

No structured customer behavior analysis

No Segmentation

All customers treated equally

Poor Delivery Optimization

Delivery issues hurt customer reviews

No Demand Forecasting

Stock issues during peak periods



SOLUTION THROUGH DATA SCIENCE WORKFLOW

This project applies an end-to-end Data Science approach:

- ETL of 100K+ real e-commerce transactions
- Exploratory Data Analysis (EDA)
- RFM-based segmentation with K-Means clustering
- Demand forecasting using Facebook Prophet
- Interactive dashboard with Streamlit



Demand Forecasting



Customer Segmentation



Interactive Dashboard



EDA and Visualization





SALES & CUSTOMER INSIGHTS

After exploration, key trends were found:

- Most popular categories: Bed Bath Table, Health & Beauty
- Most used payment method: Credit Card (73%)
- Transaction spike during Black Friday (Nov)

Favorite Products

Health & Beauty

Busy Season

November (Black Friday)

Payment Method

Credit Card dominant

Transactions

Stable, rising at year-end





CUSTOMER SEGMENTATION (RFM + CLUSTERING)



Customers are grouped based on Recency, Frequency, and Monetary. K-Means clustering is used to create 4 segments:

Champions:

Spent a lot, often, and recently

Loyal:

Routine purchases

Hibernating:

Haven't shopped in a long time

At Risk:

Low value & frequency





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SALES FORECASTING

Using Prophet Time Series to predict sales trends for main product categories.



Upward Trend

Health & Beauty



Seasonal

Early week spike



Stock Optimization

Prepare early



Recommendation

Use trend data for
campaigns



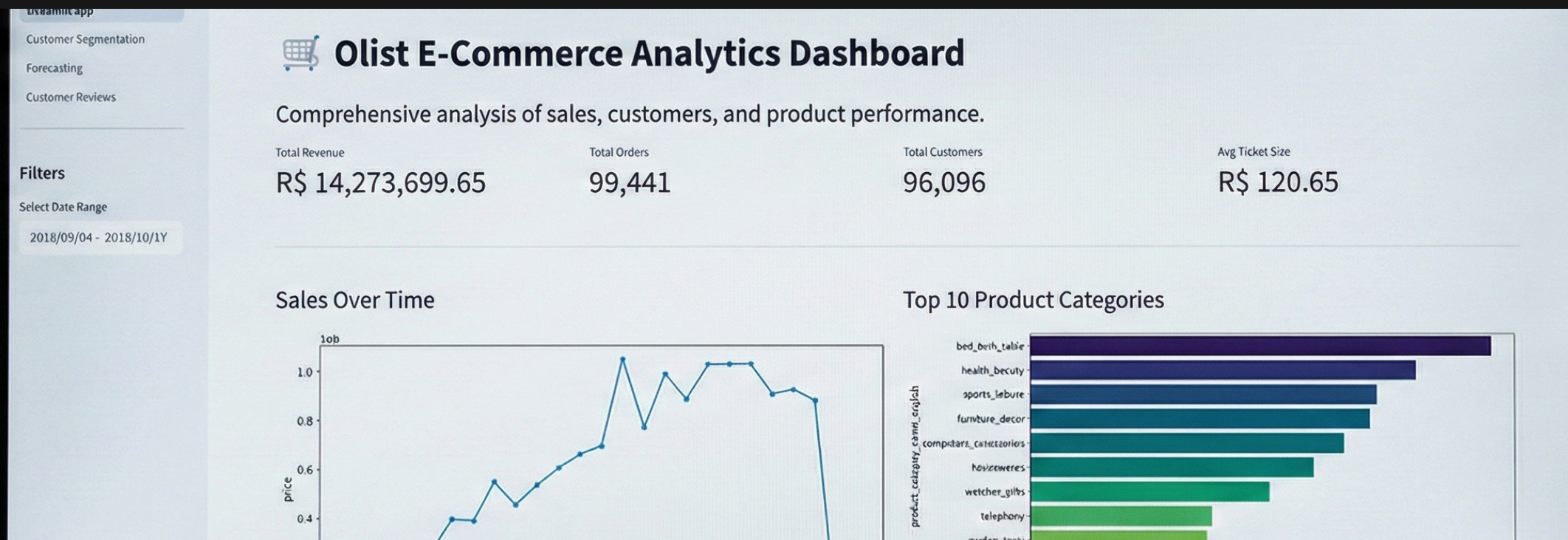
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INTERACTIVE STREAMLIT DASHBOARD

Real-time dashboard for sales and customer analysis:

- KPI Tracker
- Interactive Segmentation
- Dynamic filters
- Review analysis





KEY INSIGHTS & RECOMMENDATIONS



Insights from this analysis are used to support data-driven decision making, increase customer retention, optimize stock planning, and improve overall e-commerce service performance.

Customer Retention

Focus on loyal and champions segments

Logistics

Improve delivery performance

Stock

Increase stock during peak seasons

Feedback

Use reviews for service improvement



+62-857-9468-5605



[GITHUB.COM/MUHAMADJUWANDI](https://github.com/muhamadjuwandi)



muhamadjuwandi28@gmail.com



BOGOR, INDONESIA

LET'S TRANSFORM E-COMMERCE WITH DATA

LET'S BUILD THE FUTURE TOGETHER!

